

Ride-Hailing Trip Analysis

1. Business Problem

Uber needed better visibility into trip demand, revenue performance, and operational efficiency across locations, vehicle types, and payment methods. The lack of centralized insights into booking patterns, trip completion efficiency, and customer behavior made it difficult to optimize resource allocation, improve service quality, and maximize revenue growth.

2. Objective

Analyze booking trends, trip efficiency, revenue patterns, and customer payment behavior to identify operational improvement opportunities and support data-driven decision-making.

3. Tools & Project Links

Power BI | DAX | Excel

4. Analysis

- Performed EDA and KPI analysis
- Conducted booking and revenue trend analysis
- Performed time-based demand analysis
- Conducted pickup and drop-off location analysis
- Analyzed vehicle type performance
- Performed payment method analysis
- Conducted trip distance and duration analysis
- Evaluated trip efficiency metrics
- Built interactive Power BI dashboards using slicers, drill-throughs, and dynamic measures

5. Key Insights

- Uber completed **103.7K bookings**, generating approximately **\$1.6M revenue** with an average booking value of **\$15 per trip**
- Peak demand occurred during **8–10 AM** and **6–9 PM**, indicating strong commuter-driven ride patterns
- **UberX** generated the highest booking volume and overall revenue contribution among all vehicle categories
- Digital payments accounted for nearly **67% of total transactions**, reflecting strong customer adoption of cashless payments
- Most trips were short-distance urban rides, while long-distance trips generated significantly higher fare value
- Penn Station / Madison Square Garden emerged as the highest-demand pickup location across the network

6. Recommendations

- Optimize driver allocation during peak demand hours and high-traffic pickup zones
- Refine surge pricing strategies during evening and night-time demand periods
- Increase focus on digital payment adoption to improve operational efficiency
- Promote high-performing vehicle categories to maximize revenue contribution
- Use trip distance and location insights to improve ride distribution and pricing strategies